Justin Fox

Data Engineer

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SUMMARY

Data Engineer and analytics manager with 10+ years of experience in data analysis, reporting, and automation, and 3+ years building scalable ETL/ELT pipelines and databases. Comfortable using any modern data stack, and known for delivering reliable data systems and actionable insights using Python, SQL, and Power BI. Consistently trusted to take ownership, navigate uncertainty, and deliver high-quality outcomes.

SKILLS

Languages & Frameworks: SQL, Python, dbt, PySpark, Pandas, Scikit-learn, Requests

Data Engineering & Analytics: ETL/ELT pipelines, Delta Lake, API integration, Batch/Stream processing, Data modeling, NoSQL, A/B testing, Machine learning, Data validation

Tools & Platforms: AWS (S3, Lambda), GCP (Cloud Run, Pub/Sub, Firestore, BigQuery), Snowflake, Databricks, Power BI

EXPERIENCE

Data Strategy & Analytics Lead (Volunteer) | Jesus Said Go | Remote | August 2025 - Present

- Developed an automated WhatsApp reporting system using Deluge and API integrations to deliver personalized weekly performance stats and secure mobile verification for hundreds of users.

Principal Business Analyst / Data Administrator | San Joaquin County Probation | Stockton, CA | March 2023 - Present

- Built an automated data orchestration system using Python and SQL, creating a library of jobs running ~30 scheduled scripts to ingest, transform, and load multi-source data into a centralized SQL Server warehouse.
- Produced over 20 multi-page Power BI dashboards department-wide, increasing accessibility, efficiency, and transparency in reporting.

Business Analyst II | San Joaquin County Probation | Stockton, CA | December 2014 – March 2023

- Automated 20+ recurring reports and data extraction processes using Python, SQL, and VBA, reducing manual workload and improving accuracy.
- Developed an NLP-based machine learning model using probation case notes to predict client outcomes, achieving 89% accuracy and delivering actionable insights for management decision-making.

PROJECTS

Etsy Shop ELT Pipeline | [Github]

- Designed and deployed a fully automated cloud-native ELT pipeline that extracts incremental sales and fulfillment data from Etsy and Gelato APIs via AWS Lambda, stores raw JSON in partitioned Amazon S3, ingests into Snowflake using Snowpipe, and transforms it with dbt into a dimensional star schema with automated data quality tests and scheduled warehouse builds via Snowflake Tasks.

Website Clickstream Pipeline & Analytics | [Github]

- Designed and implemented, for a live production website for a nonprofit, a real-time clickstream analytics pipeline using JavaScript, Google Cloud Run, Pub/Sub, Firestore, BigQuery, and Looker to capture anonymized website engagement events and deliver session-level analytics.

Earthquake ETL Lakehouse Pipeline | [Github]

- An end-to-end PySpark + Delta Lake pipeline in Databricks ingesting real-time USGS earthquake API data into Bronze/Silver/Gold medallion layers with metadata logging, optimized analytic tables, and visualization outputs for reporting.

EDUCATION

California State University - Stanislaus

B.S. Mathematics | Minor in Computer Science | December 2013